

SIMATS ENGINEERING
ASSESSMENT TOOL 3
MCQ Test

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COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 20%

QUESTIONS:20

Total Marks:20

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?
 - a) Slow Start
 - b) Fast Retransmit
 - c) Fast Recovery
 - d) Tahoe Reset(BL4)(1)
2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:
 - a) 9 MSS
 - b) 16 MSS
 - c) 12 MSS
 - d) 8 MSS(BL4)(1)
3. Which field in the TCP header is used to implement flow control at the receiver side?
 - a) Sequence number
 - b) Acknowledgement number
 - c) Window size
 - d) Urgent pointer(BL4)(1)

4. Nagle's algorithm causes increased latency primarily because it:
 - a) Disables ACK piggybacking
 - b) Buffers small segments until ACK arrives
 - c) Reduces CWND on every packet
 - d) Increases SYN backlog(BL4)(1)
5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:
 - a) MTU size
 - b) CWND growth rate relative to RTT
 - c) DNS resolution time
 - d) UDP competition(BL4)(1)
6. UDP is preferred over TCP for DNS queries primarily because:
 - a) UDP supports fragmentation
 - b) DNS responses are small and connection overhead is unnecessary
 - c) TCP cannot use port 53
 - d) UDP has better error correction(BL4)(1)
7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:
 - a) Increase UDP buffer size
 - b) Switch to TCP
 - c) Implement Forward Error Correction (FEC)
 - d) Reduce frame rate to 1 fps(BL4)(1)
8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?
 - a) 1480 bytes
 - b) 1472 bytes
 - c) 1460 bytes
 - d) 1500 bytes(BL4)(1)
9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:
 - a) Eliminating IP header overhead
 - b) 0-RTT or 1-RTT connection setup
 - c) Removing encryption
 - d) Native multicast support(BL4)(1)
10. HTTP/1.1 persistent connections reduce latency by:
 - a) Enabling UDP fallback
 - b) Reusing the same TCP connection for multiple requests
 - c) Compressing HTTP headers

d) Sending requests without waiting for DNS
(BL4)(1)

11. Head-of-line blocking in HTTP/1.1 occurs when:

- a) The DNS server is slow
- b) A response blocks subsequent responses on the same connection
- c) TCP CWND drops to zero
- d) The server rejects pipelined requests

(BL4)(1)

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

- a) Application layer
- b) DNS layer
- c) TCP layer
- d) IP layer

(BL4)(1)

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

- a) 301
- b) 302
- c) 304
- d) 307

(BL4)(1)

14. In HTTP caching, the Cache-Control: no-cache directive means:

- a) Never cache this resource
- b) Validate with the server before using a cached copy
- c) Cache for 0 seconds
- d) Disable all caching headers

(BL4)(1)

15. Which DNS record type maps a domain name to an IPv6 address?

- a) A
- b) CNAME
- c) AAAA
- d) MX

(BL4)(1)

16. A DNS resolver switches from UDP to TCP when:

- a) The query type is MX
- b) The response is truncated (TC bit set)
- c) TTL exceeds 3600 s
- d) The authoritative server is unreachable

(BL4)(1)

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

- a) ARP poisoning
- b) DNS cache poisoning
- c) SYN flooding

d) BGP hijacking

(BL4)(1)

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:

a) Cache hit ratio

b) Recursive resolver query load

c) DNSSEC signature validity

d) Zone transfer frequency

(BL4)(1)

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

a) HTTP → TCP → DNS

b) DNS → TCP handshake → HTTP GET

c) TCP → DNS → HTTP

d) DNS → HTTP → TCP

(BL4)(1)

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?

a) TCP for game state, HTTP for assets

b) UDP with application-layer sequencing for game state, HTTP/3 for assets

c) HTTP/2 for all traffic

d) TCP with Nagle enabled for all traffic

(BL4)(1)

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

		responds appropriately			
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1. **Answer: c) Fast Recovery**

Reason: It reduces the congestion window (CWND) by half after detecting packet loss through duplicate ACKs, allowing faster recovery without restarting from the beginning.

2. **Answer: b) 16 MSS**

Reason: Since CWND (8 MSS) is below SSTHRESH (16 MSS), TCP is in the Slow Start phase, where the congestion window doubles after one RTT.

3. **Answer: c) Window size**

Reason: The Window Size field controls the amount of data the sender can transmit before receiving an acknowledgment, implementing flow control.

4. **Answer: b) Buffers small segments until ACK arrives**

Reason: Nagle's algorithm delays sending small packets until an ACK is received, reducing overhead but increasing latency.

5. **Answer: b) CWND growth rate relative to RTT**

Reason: A high Round Trip Time (RTT) slows the growth of the congestion window, limiting TCP throughput.

6. **Answer: b) DNS responses are small and connection overhead is unnecessary**

Reason: DNS queries are usually short, so UDP avoids the extra delay of establishing a TCP connection.

7. **Answer: c) Implement Forward Error Correction (FEC)**

Reason: FEC allows the receiver to recover lost packets without retransmission, making it suitable for real-time streaming.

8. **Answer: b) 1472 bytes**

Reason: The maximum UDP payload is $1500 - 20$ (IP header) $- 8$ (UDP header) = 1472 bytes.

9. **Answer: b) 0-RTT or 1-RTT connection setup**

Reason: QUIC reduces connection setup time by combining transport and security, enabling faster communication.

10. **Answer: b) Reusing the same TCP connection for multiple requests**

Reason: HTTP/1.1 persistent connections reduce the delay caused by repeatedly creating new TCP connections.

11. **Answer: b) A response blocks subsequent responses on the same connection**

Reason: In HTTP/1.1, responses are delivered in order, so one delayed response blocks the following responses.

12. **Answer: c) TCP layer**

Reason: HTTP/2 removes application-layer blocking, but TCP packet loss still causes Head-of-Line blocking.

13. **Answer: a) 301**

Reason: Status code **301** indicates that a resource has permanently moved and browsers should update stored links.

14. **Answer: b) Validate with the server before using a cached copy**

Reason: The browser must check with the server to ensure the cached content is still valid before using it.

15. **Answer: c) AAAA**

Reason: An AAAA record maps a domain name to an IPv6 address.

16. Answer: b) The response is truncated (TC bit set)

Reason: When a DNS response is too large for UDP, the resolver switches to TCP to receive the complete response.

17. Answer: b) DNS cache poisoning

Reason: This attack injects fake DNS responses into a resolver's cache, redirecting users to incorrect websites.

18. Answer: b) Recursive resolver query load

Reason: A low TTL causes DNS records to expire quickly, increasing the number of DNS lookups.

19. Answer: b) DNS → TCP handshake → HTTP GET

Reason: The browser first resolves the domain name, establishes a TCP connection, and then sends the HTTP GET request.

20. Answer: b) UDP with application-layer sequencing for game state, HTTP/3 for assets

Reason: UDP provides low-latency game updates, while HTTP/3 offers fast and reliable delivery of game assets even under packet loss.